Points of Attention

1. Please be careful not to damage the LIDAR mirror when installing and removing.

2.Please use the product according to the instructions in the operation guide. We are not responsible for any damage or injury caused by improper use.

3.The robot's enclosure must not be subjected to handling forces, heavy objects, or knocks.

4.To prevent damage to the equipment and violate the terms of the warranty, do not disassemble the robot body.

5.Keep sensors and LIDAR out of the way while the robot is running.

6.Before robot operation, if the travel path passes through speed bumps or potholes in the road.

7.Please edit the deceleration zone in Map Manager in advance to avoid accidents caused by robots driving through at high speeds.

8.Use of the product in extremely high (50 °C) or low (-20 °C) environments is prohibited and may cause equipment failure.

9.Please use the original rechargeable batteries and charging posts specially equipped by our company.

10.It is prohibited to use the product on wet or waterlogged floors.

11.Use the charging post carefully to avoid damage to it.

12.Do not have objects in the vicinity of the charging post to prevent the device from reaching the charging post location properly and accurately.

13.If you are not going to use the product for a long time, turn the product off.

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APP Operation Guide

1 Connecting the robot

1.1 Connection via intranet:

(1) Using the default configuration, make an intranet connection:

l Connect to the wireless WIFI of the robot through your cell phone or tablet, and the WIFI password is 12345678, as shown in Figure 1-1.

Fig. 1-1

- After connecting the robot WIFI, open the robot operation platform APP, as in Figure 1-2.

Fig. 1-2

- In the Select Robot Name field of the login screen, select the “Local Connection” option to log in normally, and the “Local Connection” option will be selected by default every time you log out to the login screen.

(2) Use a customized configuration for an intranet connection:

l Connect to the wireless WIFI of the robot itself through your cell phone or tablet, and the WIFI password is 12345678, as shown in Figure 1-1.

l Connect the robot WIFI and open the robot operation platform APP, as shown in Figure 1-2.

l Connect the robot by configuring the robot IP and port as follows: Click [Robot Configuration] to enter the device configuration interface, as shown in Figure 1-3.

Fig. 1-3

Click the [Add] button, enter the robot name, IP, port, and then click [OK], as shown in Figure 1-4.

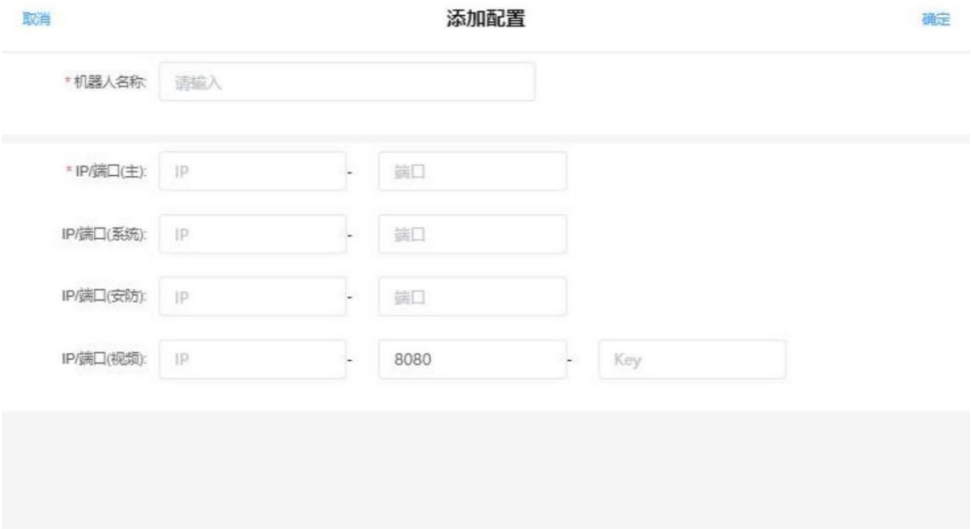

Fig. 1-4

After adding, click [Return] to exit the device configuration interface, in Figure 1-2 interface to select the name of the machine added, fill in the user name: admin; password: 123456, fill in the completion of the click on the login, enter the APP home page, as Figure 1-5

Fig. 1-5

If the APP can not connect to log in the controller after connecting to WIFI, you can try to modify the WIFI connection method to static connection/manual setting . And set the ip address to 192.168.1.210, subnet mask to 255.255.255.0, router to 192.168.1.1, dns to 8.8.8.8.To prevent address conflicts, it is best to set the ip address between 200-255.

2 Map-related operations

2.1 Mapping

1. Click on the icon button at the top of the APP homepage to bring up the remote control window, and then use the remote control handle on the right side to drive the robot to a certain position.

2. Click the icon button, as shown in Figure 2-1.

Fig. 2-1

Enter into the [Select Map] function interface, click the upper right corner of the [Build Map], as shown in Figure 2-2.

Fig. 2-2

Jump to the [Create Map] function interface, as shown in Figure 2-3.

Fig. 2-3

3. Click [start building], will enter Figure 2-4 map initialization interface, the peripheral red points on the map for the laser point data; start scanning the robot's starting position will be recorded as the origin of the map, and become the map of the origin initial point.

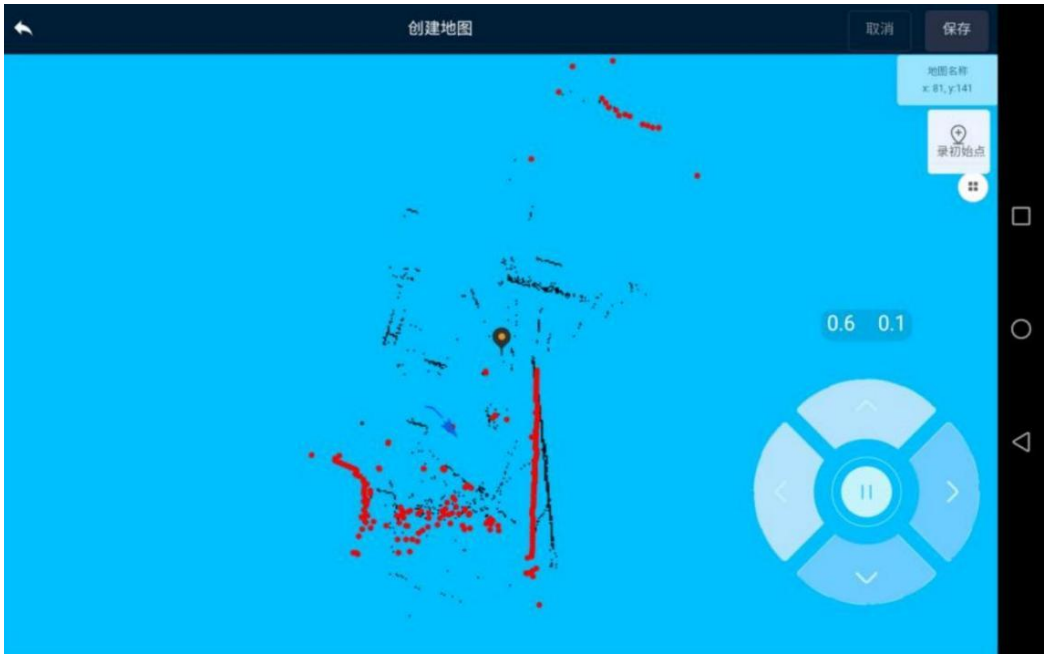

Fig. 2-4

4. Through the remote control on the interface to control the robot to walk, the robot will automatically generate a map of the walking environment through the laser interface when the robot has finished walking the environment to be scanned, click the stop button in the middle of the remote control interface to make the robot stop, and then click the upper-right corner of the interface to [Save] button, and input the name of the map (letters and numbers), click to confirm (Figure 2-5), the system will optimize and save the scanned maps. The system will optimize and save the scanned map. After the map is finished, it will automatically jump to the [Select Map] function interface.

Fig. 2-5

5. In the constructed map, dark blue indicates obstacles fed back from laser scanning, light blue indicates areas not scanned by the laser, white indicates no obstacles, and gray lines indicate road edges fed back from laser scanning. Note: Curbs are only present when using the 3D laser to build a map, as shown in Figure 2-6.

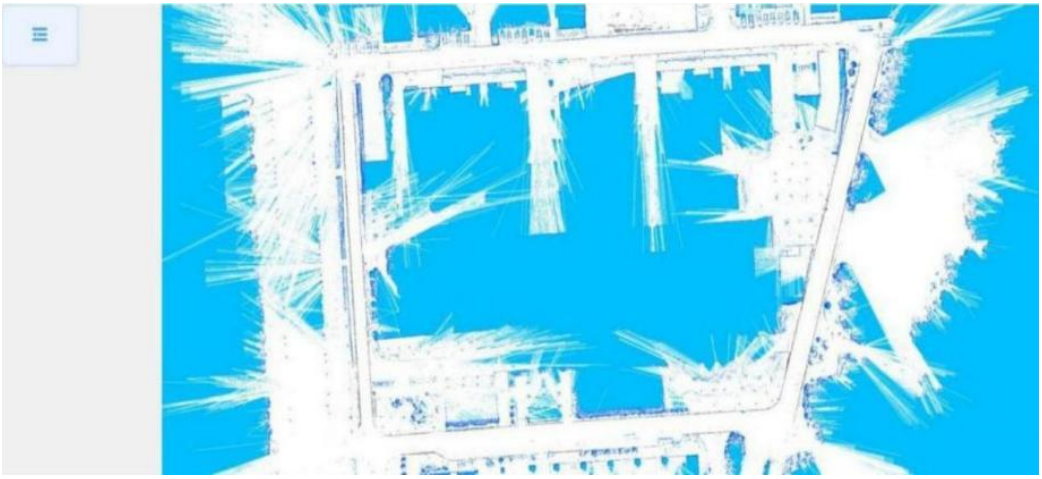

Fig. 2-6

2.2 Mapping pointst

The function of map recording can record points during the process of building a map, and the recorded points can be used for navigation, positioning, and drawing tracks, which is only applicable to 3D laser at present, and the main operation steps are as follows. 1:

1. In the APP home page, enter the [Select Map] interface, click the [Build Map] button in the upper right corner, and jump to the [Create Map] interface. 2. Click [Start Building], and then click [Create Map].

2. Click [Start Map Building], use the remote control on the interface to control the robot to walk, and stop the robot at the place where you need to record points. 3. Click [Record Initial Points], and then click [Record Initial Points].

3. Click [Record Initial Point], input the point name, click OK, and the recording will be successful, as shown in Figure 2-7. When you select the map after the map is built, you can see the recording point.

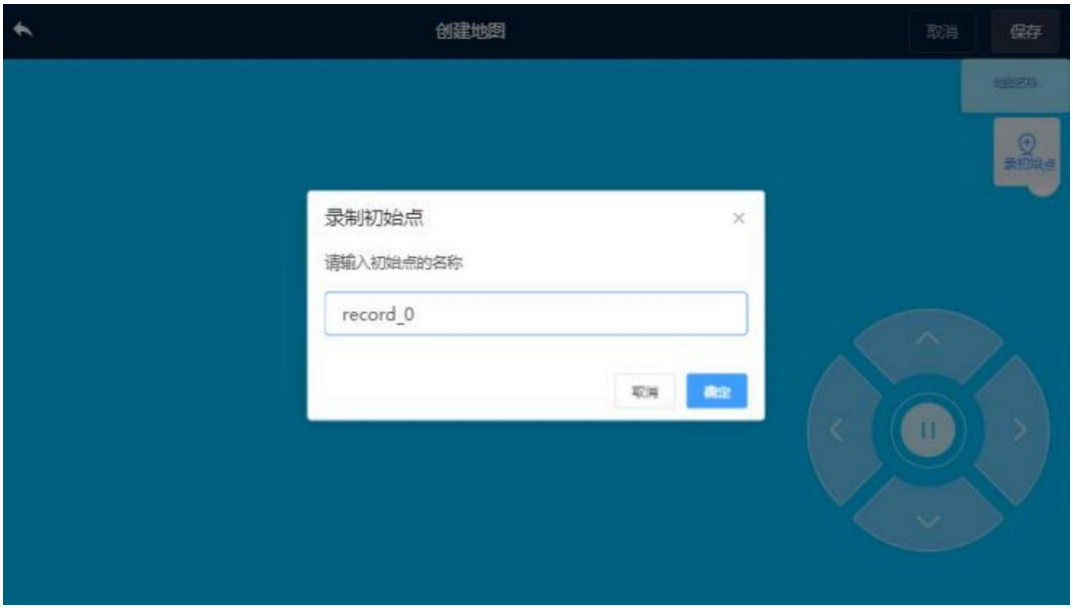

Fig. 2-7

4.It will be more accurate to draw the track by recording the point connection, so if you want to draw a more reasonable track, you have to complete the track drawing by remote controlling the car to the corresponding position to record the points after the

drawing is finished, in order to save the time and avoid repeating the operation, we can stop the car at the place where we need to record the points to record the points, and then continue to move the car remotely to build the track when the point recording is finished. Note: This function is only effective for 3D laser, and the recorded points can be connected with each other normally, the precautions are described in detail in the function of hand-drawn track.

2.3 Continued mapping (not yet open)

(Note: This function is limited to 2D environment only!)

Continuing to build a map means that you can continue to build a map based on the original map. Before continuing to build a map, you need to move the robot to the initial position when you continue to build a map, and the direction of the laser and the cart with the original map can be initialized.

2.4 Rules and recommendations for building maps

1. The construction of the map must go through a complete loop, i.e., the robot's travel path should form a complete closed loop when building the map.
2. In environments with many forks and loops, close small loops before large ones.
3. Try not to rotate the starting point of the build.
4. The end point is chosen to be on the previously traveled path, with as little rotation as possible at the end point. If the end point is near the start point, then the end point can be chosen to be beyond the start point.
5. The operator stands behind the robot as far as possible.
6. Other relevant personnel should try not to stand in front of the robot when building the map, otherwise it will be easily added to the map as a fixed obstacle when building the map.
7. Try to rotate as slowly as possible in turning places, and try to walk in a straight line if you can, or walk in a straight line and rotate in place if you have to walk in an arc.

8. If you are building in 3D, you need to change the environment parameters 0 (indoor) and 1 (outdoor) in the system update.

2.5 Editing a Virtual Wall

Editing the map is mainly for editing and modifying the built map, among which the Edit Virtual Wall function is used to manage the map functional areas, so that the robot can navigate to the corresponding functional areas and perform the corresponding operations. There are six types of virtual wall function areas:

l Where there are walls or objects in the map that cannot be recognized by lasers such as glass walls, translucent light, etc., where there are fragile objects, steps, or or where you don't want the robot to pass through, set up a virtual barrier so the robot won't pass through the area.

2 Ramp: the robot will slow down before navigating to the area.

3 Intersection: The robot will detect traffic lights when traveling to this area and stop at the red light and proceed at the green light. Note: A separate traffic light detection module is required.

4 Deceleration zone: The robot decelerates when it navigates to this zone.

5 No Obstacle Avoidance Zone: When the robot navigates to this zone, if it encounters an obstacle, the robot stops but does not perform obstacle avoidance until it is safe to proceed ahead.

6 Obstacle avoidance zone: when the robot navigates to the zone, it will consider the zone as an obstacle and perform an obstacle avoidance detour.

1.Steps: In the APP home page, enter the [Select Map] function interface, click the right side [Edit] button, as Figure 2-8.

2.

Fig. 2-8

Enter the [Edit Map] function interface, as shown in Figure 2-9.

Fig. 2-9

Select [Edit Virtual Wall] to enter the main interface of Edit Virtual Wall, as in Figure 2-10.

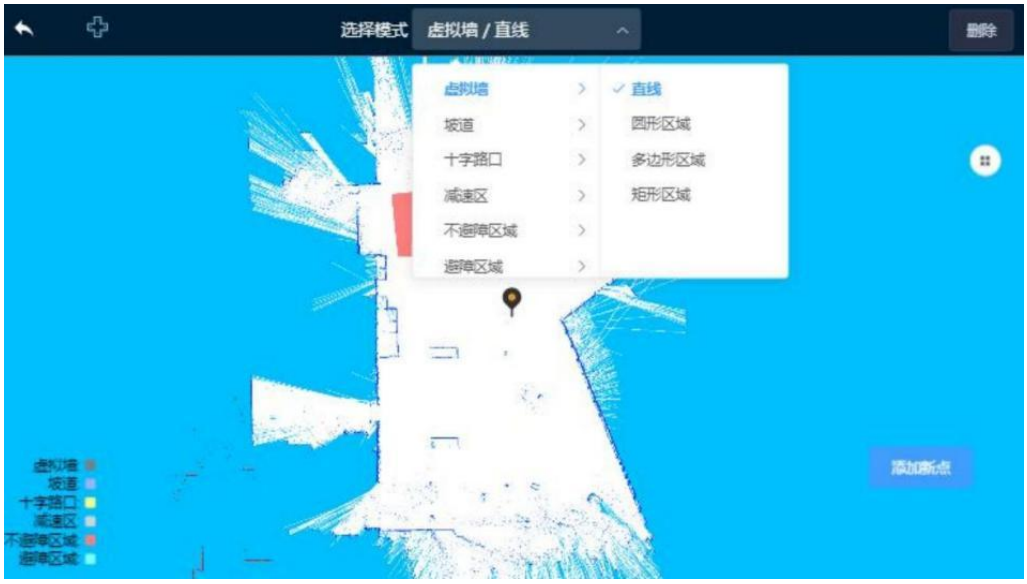

Fig. 2-10

In the upper part, you can select the mode, including virtual wall, ramp, intersection, deceleration zone, non-obstacle avoidance zone, obstacle avoidance zone, and there are four types of virtual wall in the selection of virtual wall mode: straight line, polygon, rectangle, and circle:

●Straight line: Determine the starting and ending points of the straight line on the map. After selecting a straight line, drag the map to a suitable location so that the black pin is located at the starting point of the virtual wall to be drawn, click on the right side of the [Add New Point] button, drag the map so that the pin is located at the end point of the virtual wall to be drawn, click on the right side of the [Connecting Points to] button, you can draw a straight line virtual wall as in Figure 2-11 If the virtual wall meets the requirements, click on the button “Finish” to save the virtual wall.

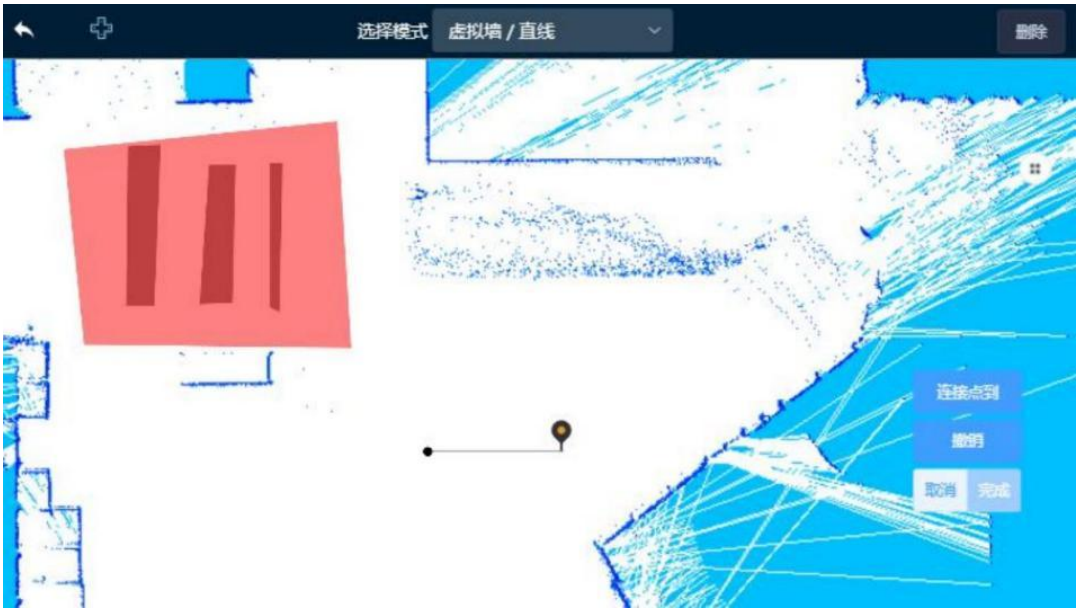

Fig. 2-11

- Polygons: Simply identify multiple vertices of a polygon on the map;

l Rectangle: Identify the two diagonal points of the rectangle on the map;

l Circle:Determine the center of the circle on the map and drag the map to determine the radius of the circle.

2. When choosing the mode of ramp, move the red dot in the figure to the position where you need to draw the virtual wall, click on the right side to add a new point, and determine the polygon's multiple vertices on the map, and then click on the [Finish] button after the connection is completed, you can save the virtual wall (as in the dark gray area in Fig. 2-12); the same operation is performed for the mode of intersection (as in the yellow area in Fig. 2-12) and deceleration zone (as in the light gray area in Fig. 2-12). The operation is the same for the mode of intersection (yellow area in Figure 2-12) and deceleration zone (light gray area in Figure 2-12).

Fig. 2-12

3.Delete Virtual Wall: If you want to delete a virtual wall, just click on the virtual wall in the figure to select it (e.g. Figure 2-13 Virtual Wall is green when selected), and then click on the Delete button in the upper right corner to delete the virtual wall.

Fig. 2-13

2.6 Editing obstacles

Due to the large number of people and cars in the outdoor situation when building the map, there will be map noise or some obstacles can't be recognized, so we can clear the noise on the map or add obstacles manually.

In the APP home page, click to enter the [Select Map] function interface, click the right [Edit] button to enter the [Edit Map] function interface, select [Edit Obstacle], then you can enter the [Edit Obstacle] main interface. Figure 2-14 below.

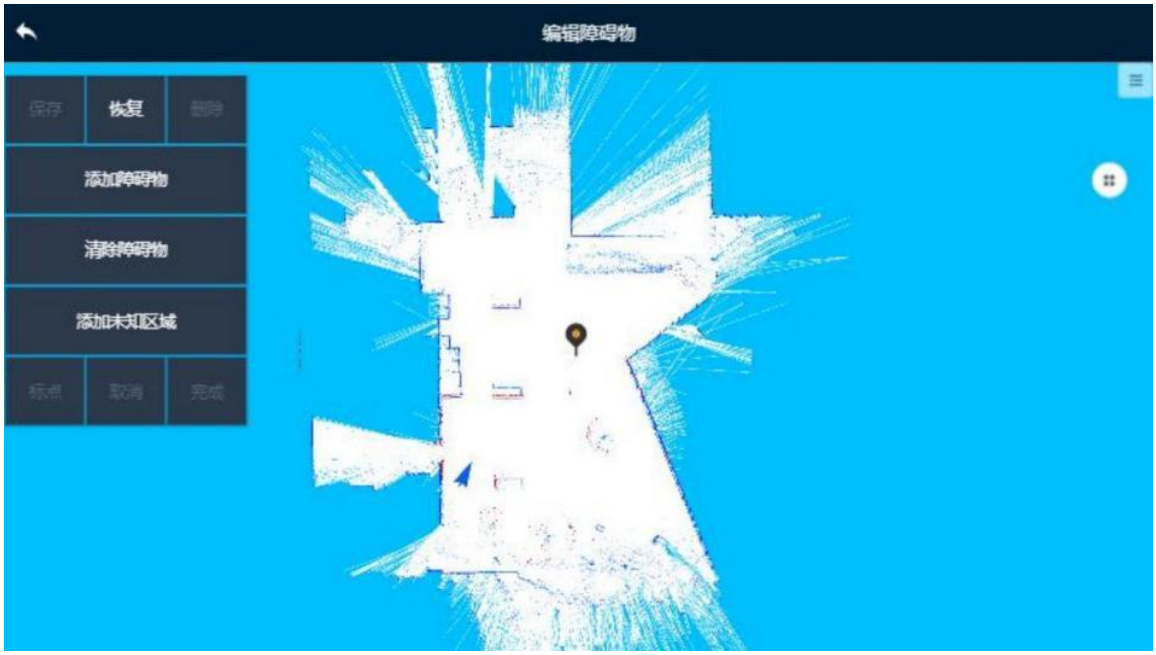

Fig. 2-14

The left test option allows you to manually add obstacles, remove obstacles, and add unknown areas. If you accidentally add or remove a map in the wrong location, you can click the [Recovery] button to restore the map to the original.

Below is an example of the steps to add an obstacle:

(1) Click Add Obstacle, and the dot function will light up.

(2) Move the black pin to the place where the obstacle is not scanned and click [Mark Point], then move the pin to the area where the obstacle needs to be added and click [Mark Point] several times.

(3) Click [Finish] button after marking is completed, and the added obstacle areas will be displayed in the list on the right, and multiple obstacle areas can be added. Note: Currently does not support the simultaneous addition of obstacles and obstacle removal, you need to add obstacles to save the first, in the removal of obstacles.
(4) Click the [Save] button to display the added obstacles on the map.

After using the 3D laser to build the map, there will be a road edge in the map by default, and you can use the Edit Obstacle function to remove the road edge. Note: The road edges are for better display and control of the path the robot is traveling.

3 Navigation-related operations

After building and editing the map, the robot can start to perform navigation-related operations. Navigation related operations mainly include position initialization, point recording, path recording, mission running, hand-drawn points, hand-drawn tracks, hand-drawn paths, new missions, etc. All these operations can only be performed by selecting the desired map after entering the corresponding menu.

- Position initialization: enables the robot to be correctly positioned in the map.

- Record Points: Record the orientation and coordinate points of the robot's current location.

- Hand-drawn points: By dragging the map you can artificially select any point and any orientation within the map.

- Hand-drawn tracks: add tracks to the navigation system to aid in navigation path generation.

- Record Path: Record the path that the robot actually travels.

- Hand-drawn paths: Drag and drop any two points on the map to create a path between them (no obstacles in between).

- New Task: Combine drawn paths, points, areas, and parking spaces into a single task by selecting them in a combination, or execute them individually.

➢ Running Tasks: In the Running Tasks screen, you can run combination tasks and temporary tasks (including running points, paths, and zones).

➢ Running combination task: The robot will automatically navigate and follow a path through one or more recorded coordinate points or a combination of multiple points and multiple paths in a selected order.

➢1 Running Points: The robot will walk according to the actual recorded points or hand-drawn points .

➢ Run Path: The robot will follow the actual recorded path above or the hand-drawn path.

➢ Running area: the robot will walk in the area .

3.1Position initialization

The robot is correctly localized so that it can perform navigation tasks correctly.

In the APP home page, enter into the [Select Map] function interface, after selecting a suitable map, see whether the robot position and orientation are consistent with its position and orientation in the actual map, if they are consistent, then you can directly carry out the next step of other operations, if they do not match, you need to help the robot repositioning, click the icon on the home page to enter into the [Position Initialization] main interface for the initialization of the robot's position operation, as shown in Figure 3-1.

Fig. 3-1

There are three types of position initialization:

- Fixed-point initialization: (Recommended) Drive the robot to the location of an existing recording point in the map, adjust the robot position in the same direction as the recording point, click the [Point Initialization] button, and then select the name of the recording point in the list on the left side, and then click the [Point Initialization] on the right side of the top to locate the point, Note: All recording points and origin points can be used as initial points for fixed-point initialization;

- Manual Initialization: Tap [Manual Initialization] on the right side of the top, move and rotate the map, move the robot to the actual position of the robot in the map, rotate the map until the red laser dots around it match the map frame, and then tap the [Done] button until you see the laser match the map frame.

3.2Position initialization

(1)Remote control of the robot to the appropriate position and orientation after accurate localization

(2)In the APP home page, click the “ ” icon in the lower left corner, select [Point Management] to enter the main interface of point management, click [New] in the upper right corner to pop up the prompt box as in Figure 3-2, select [Record Point], enter the point name and select the point type;

Fig. 3-2

(3) Type of recording point:

- Charging Stake: Used for charging the robot (the charging stake points must be entered by means of point recording);
- Navigation points: normal navigation points;
- Inspection points: normal inspection points;

(4) Click [OK], the system will record the current position and orientation of the robot as point position information;

3.3 Hand-painted points

You can directly select suitable points and point orientation in the passable area on the map for point recording, and you don't need to remotely control the robot to a specific point. 1.

1. In the home page of APP, click the “” icon at the bottom left corner, select ’Point Management” to enter the main interface of point management, the list on the left shows the list of points that have been added to the map, click the name of the point in the list, and the point will be highlighted in the map. 2. Click the New button, a prompt box will pop up in Figure 3-3, enter the appropriate point name and point orientation.

2. Click the New button to bring up the prompt box in Figure 3-3, enter the appropriate point name and point type, and confirm.

Fig 3-3

3. Drag the zoom map so that the pegs in the position of the need to record points, click the [Confirm] button to confirm the location of the point, and then drag the map, just confirmed that there will be an arrow on the point for the display of the point's orientation, dragging the map to change the direction of the arrow, after confirming the direction, and then click the [Confirm], the point will be saved in the system, such as Figure 3-4.

Fig 3-4

3.4 Hand-drawn tracks

In order to keep the robot on the specified track or let the robot walk in the middle or at the side of the road when running point in outdoor or open field, it is necessary to draw a reasonable track in advance to assist the navigation of the running point path generation before performing the running point task. In the process of building the map, the robot will automatically record the walking path of the robot and generate the track according to the path, if the track does not meet the requirements, it is necessary to redraw the track by hand, and the redrawn track will automatically display the original track after it is saved successfully.

1. In the APP home page, click the “ ” icon in the lower left corner, select [Hand-drawn Track], enter the main page of hand-drawn track, the upper left side shows the function buttons, select the hand-drawn track, the hand-drawn track of the map will be displayed in the right map. 2. Click the [Rebuild Track] in the upper right corner, the hand-drawn track will be displayed in the right map.

2. Click [Rebuild] in the upper right corner, after the prompt box pops up, click [Confirm] button to enter the interface of hand-drawn track. 3.

3. You can add tracks through the map recording points. Before adding a hand-drawn track, you need to add the corresponding recording points on the map in order (Figure 3-5).

Fig 3-5

Then select the appropriate recording point name and click [Add]. Note: In order to solve the problem of repetitive tracks when drawing tracks by hand, we added the function of [Reset Start Point], its function is to reset the start position of a certain point, and the reset point will be used as the start point to connect to another point, so that you don't need to draw repetitive tracks to connect. 4.

4. In the hand-drawn track interface, click the point list, and click [Add Point] once every time you select a navigation point to connect all navigation points. 5.

5. After drawing, click [Save] to save the track, and the track will automatically overwrite the original track in the system.

6. If there is a right-angled corner in the map, you need to record a point at each end of the corner to connect it, for example, in Figure 3-6, point 2-3, point 5-6, because a purely straight line connecting the corner is not conducive to the walking of the Ackermann chassis, which will affect the effect of the robot's driving, we need to change the curvature of this connecting track in the Edit Track interface, so that the track will become more rounded and smooth, which is suitable for the car to walk around the corner normally. We need to change the curvature of this track in the Edit Track interface to make it more rounded and suitable for the normal traveling of the car.

Fig 3-6

Edit hand-drawn tracks: mainly edit the curvature degree of the line segment, set the single or double nature of the track, add or delete tracks.

1. Edit the curvature of the track: Click Edit in the interface of the hand-drawn track to enter the interface of editing the track, select the track list on the left side, click the track list, select the name of the specific line segment at the bottom of the track list, the selected line segment will be displayed in red in the map on the right side, select the track segment you want to edit, and slide the slider to adjust the curvature size of the track segment, which will make the track more curvature;

2. Edit the one-way and two-way of the track: in the track list interface, there are one-way and two-way choices: choosing one-way means that the track can only reach the end point from the start point, for example, it can only reach point 06 from point 05, as shown in Fig. 3-7; choosing two-way means that the two points of the track can be both the start and the end points, for example, it can reach point 02 from point 01, and it can also reach point 01 from point 02.

Fig 3-7

3.Add or delete part of the track: in the track editing interface, click the point list, in the drop-down box of the point list, select a point, click [Reset Start Point], you can reset a certain point as the start point, then select a point that needs to be connected, click [Add Point], then you can add a section of the track; in the track editing interface, click the track list, in the drop-down box of the track list, select a section of track that needs to be deleted, click [Delete], then the selected track can be deleted. In the track editing interface, click the track list, in the drop-down box of the track list, select a section of track to be deleted, click [Delete], and the selected track can be deleted.

3.5 Record Path

1.In the APP home page, click the “ ” icon in the lower left corner, click [Path Management], enter the [Path Management] function interface, click [New] in the upper right corner, select [Record Path] in the pop-up box and enter the path name, click [OK] button.

2.Click [Start] to remotely control the robot to walk in the map, and the system automatically records and optimizes the path that the robot has really taken.

3. Click the [Save] button to save the path.

Fig 3-8

3.6 Hand-drawn paths

Without the need for a remote-controlled robot, multiple points are manually selected on a map, and the system automatically connects the selected points in a straight line to form a path;

1.In the APP home page, click the “ ” icon in the lower left corner, click [Path Management], enter the [Path Management] function interface, the list on the left side shows the paths that have been established.

2.Click the [New] button in the upper right corner, select [Hand-drawn Path] in the pop -up box, enter the path name and confirm, then enter the path drawing interface.

3. Through the interface of the pins, drag the map to select multiple points, the system will be two points with a straight line between the connection as a path.

4. If the line between two points passes through a solid wall or a solid obstacle, it will indicate that “the path is not safe” and need to be re-selected.

5. After the drawing is completed, save it through the [Save] button at the bottom right of the map, as shown in Figure 3-9.

6. The left side will show the drag bar that draws the corresponding path name and controls the arc of the path.

7. Click the right “ ” icon in this interface, respectively check the right side of the options, you can show and hide all the points, paths, virtual walls, tracks and so on.

Fig 3-9

3.7 Creation of tasks

If you need to execute multiple runpoints or paths consecutively, or automate certain runpoints or paths multiple times, you need to create the corresponding tasks first.

1. In the APP home page, click on the “ ” icon in the lower left corner to enter the [Combined Task] function interface, click on the upper right corner of the [New], enter the task name, enter the new task interface, click on the drop-down box at the top, it will be displayed in the left list of all the points, paths, areas and parking spaces previously created, check the points/paths/areas/parking spaces that need to be added. Check the points/paths/areas/parking spaces you want to add as shown in Figure 3-10.

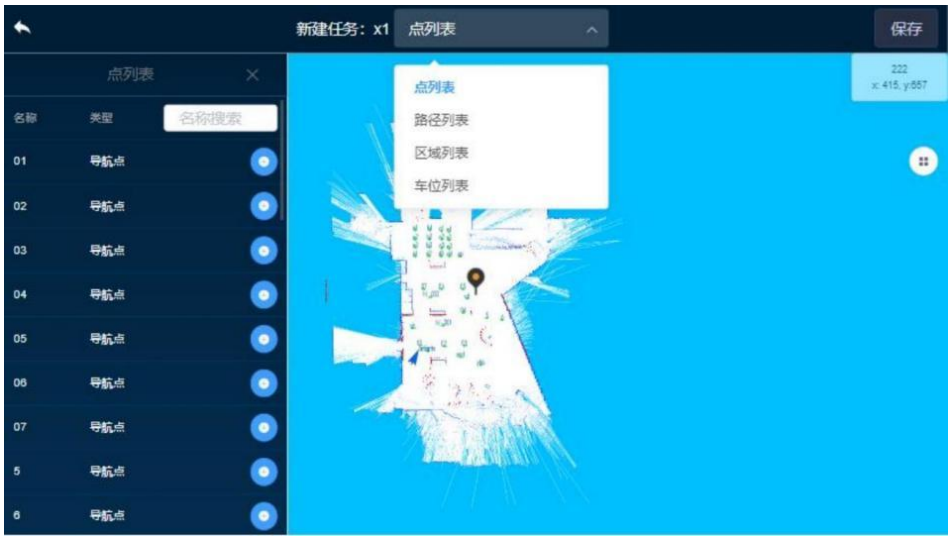

Fig 3-10

Click on the right side of the list (Add) button, a pop-up window will appear (Figure 3-11) which can be added to set the robot to run the point or path when some of the action, set the action and parameters, click on the [Add Action], the selected action and parameters will be displayed in the list below, click on the [OK] button, the point/path/area/parking space as well as the corresponding action will be added to complete the click on the pop-up box on the upper right corner of the Click “x” to close the pop-up window, you can view the points/paths/areas/parking spaces that have been selected, after viewing, click [Add] and you can continue to add points/paths/areas/parking spaces, after adding, click the upper right corner of the [Save] button, a task has been created successfully. When there are more points/paths/areas/parking spaces, slide the list area to show the hidden data.

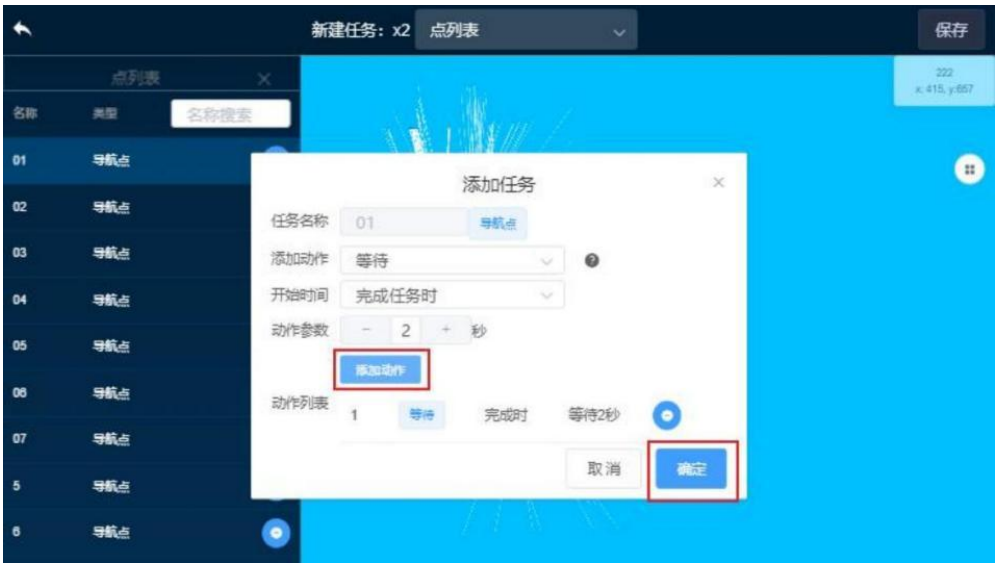

Fig 3-11

3.8 Running tasks

3.8.1 Running tasks

1. In the home page of APP, click [Start Task] to enter the combined task interface. Click [View] in the upper right corner of the task to view the points and paths selected in the task, as shown in Figure 3-12.

Fig 3-12

2. If you click [View] in the upper-right corner of the task, you will be redirected to the detailed interface of the task, and you can delete the task by clicking [Delete], as shown in Figure 3-13.

Fig 3-13

3.In the [Combined Tasks] function interface, click the “Infinite Cycle” sliding box, when the color changes to green, it is an infinite cycle state, and the robot will cycle through the selected tasks, or click the plus button next to the sliding box, you can perform the task any number of times, the default is 1 time, and the maximum is 50 times.

4. Click [Start Task! The robot will follow the created combination of tasks in order to complete the given task. If the remaining number of runs is greater than 1, the robot will automatically enter the run state again and execute it until the set number of runs has been completed.

5. During the execution of the task, the [End] [Pause] [Continue] buttons can be used to end, suspend and continue the operation of the task, as shown in Figure 3-14.

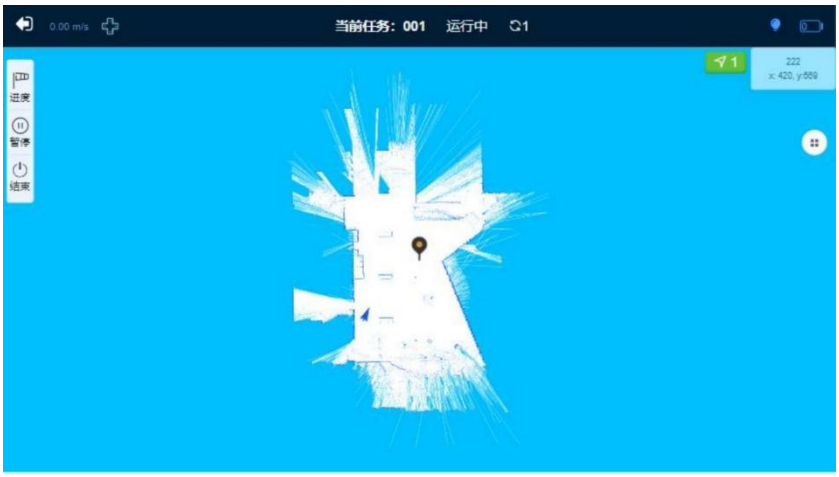

Fig 3-14

3.8 Running tasks

Let the robot walk along the selected route.

In the APP homepage, click [Start Task] to enter the combined task interface. Click [Temporary Task], in the temporary task interface, select the list of routes through the top drop-down box. As in Figure 3-15.

Fig 3-15

1. Run path: Select the specific path name in the left list, click the [Start] button on the upper right, the robot will automatically plan the path movement to the starting point of the path, and then walk along the selected path to the end of the path. 2.

2. During the running process, you can control the robot's movement status through the [Pause], [Continue] and [End] buttons. 3.

3. Delete the path: In the APP home page, click the “ ” icon in the lower left corner, click [Path Management], enter the path management function interface, click the delete button in the left list, then the path can be deleted from the system.

3.8.3 Running point

After the robot is correctly positioned and the track is plotted, it can execute the running point task.

In the APP home page, click [Start Task] to enter the combined task interface. Click [Temporary Task], and in the temporary task interface, select the point list through the top drop-down box.

1.Run point: Select the specific point name, click the [Start] button on the upper right, the robot will automatically move from the current position to the selected point.

2. The robot motion status can be controlled by [Pause], [Continue] and [End] buttons during the running point.

3. Delete Points: In the APP homepage, click the “ ” icon in the lower left corner, click [Point Management], enter the [Point Management] function interface, select the specific point name, click the [Delete] button, then the point can be deleted from the system.

3.9 Timed Tasks

3.9.1 Creating timed tasks

In the APP home page, click the “ ” icon in the lower left corner, click [Timed Tasks], enter the [Timed Tasks] function interface, you can view the list of timed tasks that have been created, as shown in Figure 3-16.

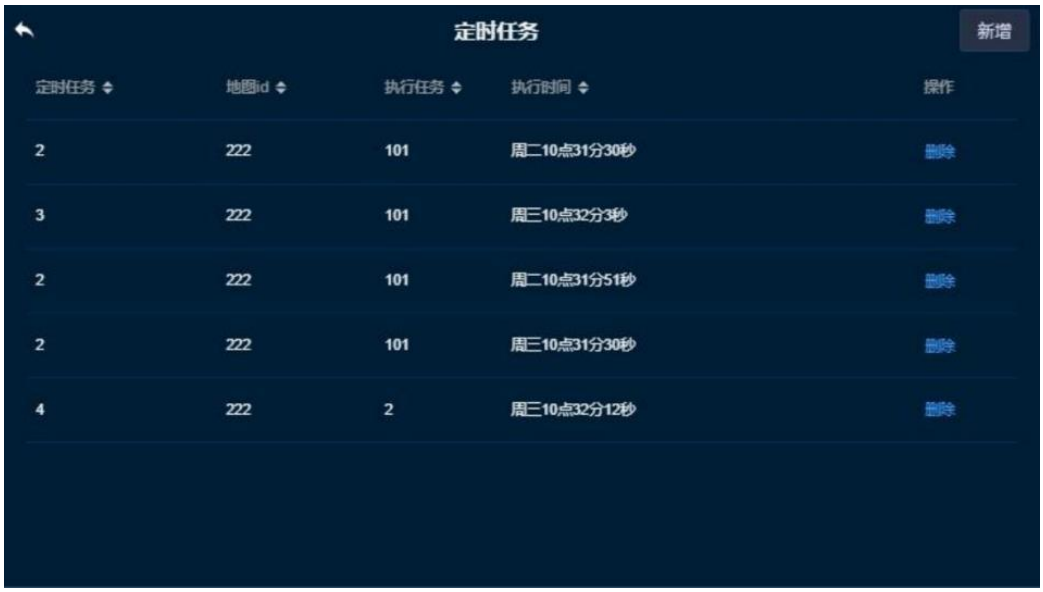

Fig 3-16

Click the upper right corner of the [New], enter the [New Timed Task] function interface, fill in the form as needed, such as Figure 3-17, click [OK] to save a new timed task.

Cyclic or not: the task is repeated weekly at the designated time;

Interrupt Task: When executing a timed task, you can automatically stop the task that is being executed manually and proceed directly to the timed task.

Fig 3-17

3.9.2 Deleting timed tasks

Enter the [Timed Tasks] function interface, you can view the list of created timed tasks, click the [Delete] button on the right side of the list to delete a timed task.

3.10 Mission reports

In the APP home page, click the “ ” icon in the lower left corner, click [Task Report], enter the [Task Report] function interface, you can view the list of task reports, such as Figure 3-18

Fig 3-18

Click on the table row to jump to the View Task Report Details, as shown in Figure 3-19.

Fig 3-19

4 System-related

4.1 Parameter Configuration

You can set the robot setting parameters online. In the APP home page, click the “back” icon in the lower left corner and click [Navigation Configuration] to enter the navigation configuration function interface, as shown in Figure 4-1.

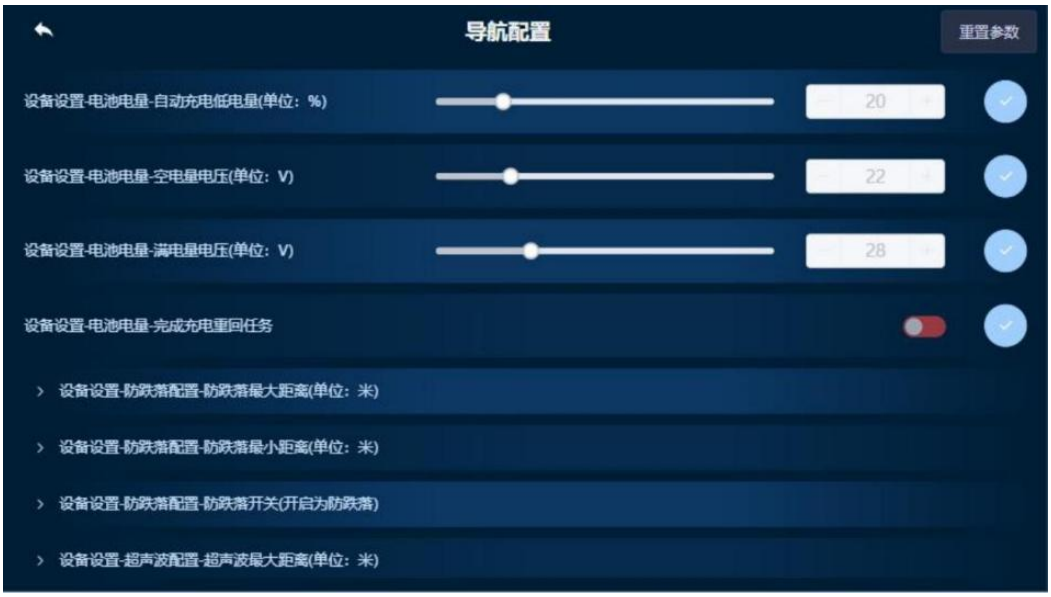

Fig 4-1

1.Click on the table row to enter the editing state, modify the parameters and then click on the right side of the “ √ ” button, the modified parameters will be uploaded and saved;
2.Reset Parameters: Resets all parameter values to factory settings;
3. Valid and upper/lower limit: bool value variable, 1 is valid, e.g. IR switch is set to 1 for turning on the IR device, 0 for turning off. The upper and lower limits mainly set the setting range of the parameter, now it is mainly used for the speed setting of navigation and following, setting 0,1,2 means low, medium and high speed level respectively.

4.2 Record package management

In the APP home page, click the “ ” icon in the lower left corner, click [Recorded Package Management], and enter the [Recorded Package Management] function interface, as in Figure 4-2, in this interface, you can view the information of the recorded package name, as well as you can delete the discarded packages.

Fig 4-2

4.3 System information

In the APP home page, click the “ ” icon in the lower left corner, click [System Information], and enter the [System Information] function interface, in which you can check the APP version, firmware version, firmware upgrading, Log package downloading, synchronizing time, checking the system status, restoring factory settings and other functions. As in Figure 4-3

1.Firmware Upgrade: In this column, you can check the APP version, firmware version information, if you can't show the latest information after updating the version, you can click [Refresh] button. In this column, you can upgrade the firmware of the robot.

2.Log package download: When the robot has a problem, you can download the log package for developers to troubleshoot the problem.

3.Synchronized time: synchronizes robot time with device time.

4.System Status: View CPU temperature, CPU utilization, memory, disk information.

5. Restore Factory Settings: After clicking Restore Factory Settings, all data will be erased and restored to the factory state, please use this function with caution.

Fig 4-3

5 Status display and control

5.1 Controlling robot movement

In the APP home page (Figure 5-1), click the “” icon at the top of the interface, a remote control window will pop up at the bottom right corner to control the robot's movement, by clicking the direction buttons to control the robot's movement back and forth, left and right, the center button is the emergency stop button, you can control the robot to stop the movement, each additional click on the direction buttons will make the Each additional click on the direction button will cause the robot's movement speed level to increase or decrease by one level.

Each additional click on the direction buttons will cause the robot's movement speed level to increase or decrease by one level. If you want to close the robot movement window, you can click the “” icon at the top of the interface again to close the remote control.

Fig 5-1

Remote control speed:

In the APP home page (as in Figure 5-1) , click the “ ‘ icon button to enter into the [Function Setting] interface, click the ’ ” icon in the upper right corner to expand the hidden setting area, as in Figure 5-2

Fig 5-2

The remote control speed is adjusted to high speed, and the maximum traveling speed can reach 1.2 m/s. The remote control speed is adjusted to low speed, and the maximum traveling speed can reach 0.6 m/s. The default is low speed.

Remote Control Type:

In the hidden setting area, the remote control type is set to joystick, you can switch the remote control to joystick type, after switching the remote control as in Figure 5-3 (the default remote control is the arrow remote control in the form of up and down, left and right).

Fig 5-3

5.2 Robot information display

In the APP home page (as in Figure 5-1), click the “ ” icon at the top of the interface, you can get the robot's moving speed, display steering angle, power and other information, as in Figure 5-4 below:

Fig 5-4

- 本地连接: Current robot name
- 0.00 m/s: Current robot speed;
- 0 rad/s: Current robot rotation angle.
- 2ms: Real-time display of network signals.
- E2 : Show/Hide Pan/Tilt Control Remote
- 底盘状态:Check the status of the robot's chassis.
- 录包: Recording package icon for recording map, navigation, and location packages.

5.3 Chassis condition

In the APP home page (Figure 5-1), click the “ ” icon at the top of the interface, and click the [Chassis Status] button in the pop-up box, which can display the robot's GPS, each infrared status and distance, ultrasonic status and distance, and voltage information, as follows Figure 5-5

Fig 5-5

5.4 Recording Packages

In the APP home page, click the “ ‘ icon at the top of the interface, and then click the ’ ” icon in the pop-up box to bring up the interface of package recording, as shown in Figure 5-6.

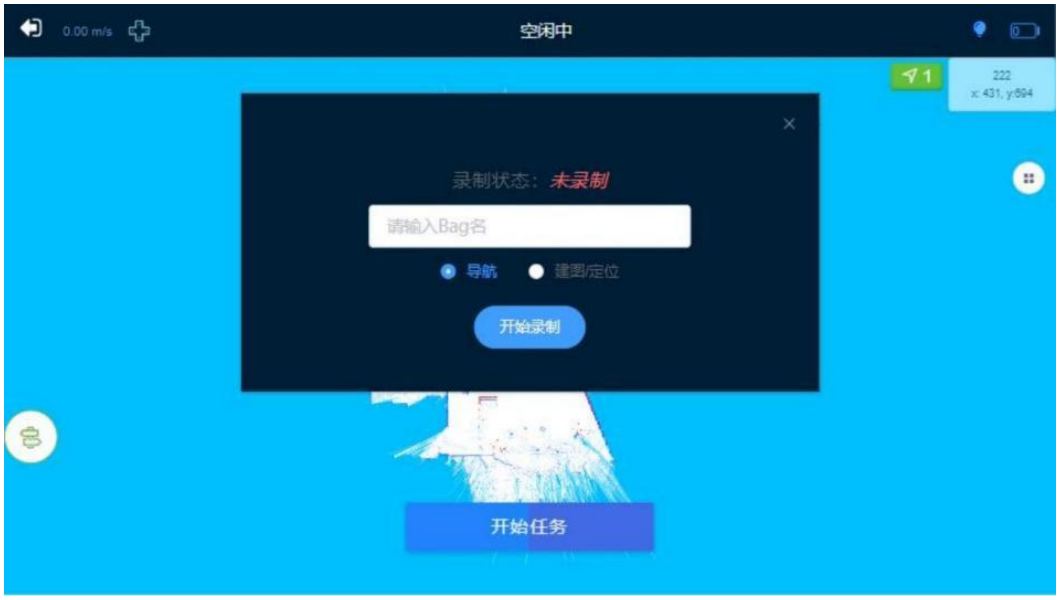

Fig 5-6

The function of recording package is to find the problem, you can send the recorded package to the staff, it is convenient for the staff to find the problem quickly, when recording the package, you need to click on the [start to build the map] before the recording package; when the recording icon is red, it means that the package is being recorded; the main types of recording packages are:

- Navigation mode: maps and navigation packages are all recorded.
- Map building/localization: recording when inaccurate positioning of the robot occurs.

5.5 Head Control

In the APP home page (Figure 5-1), click the “ ‘ icon at the top of the interface, click the [PTZ Control] button in the pop-up box, and a PTZ remote control will pop up at the bottom right corner, and you can control the PTZ up and down, left and right by clicking the direction buttons, and the center button is to stop the rotation, and the ’ ” button is the reset button at the bottom right corner. The center button is to stop the rotation, and the bottom right corner “ ” is the gimbal reset button. If you want to

close the Gimbal Control window, you can click the [Gimbal Control] button again to close the remote control, as shown in Figure 5-7.

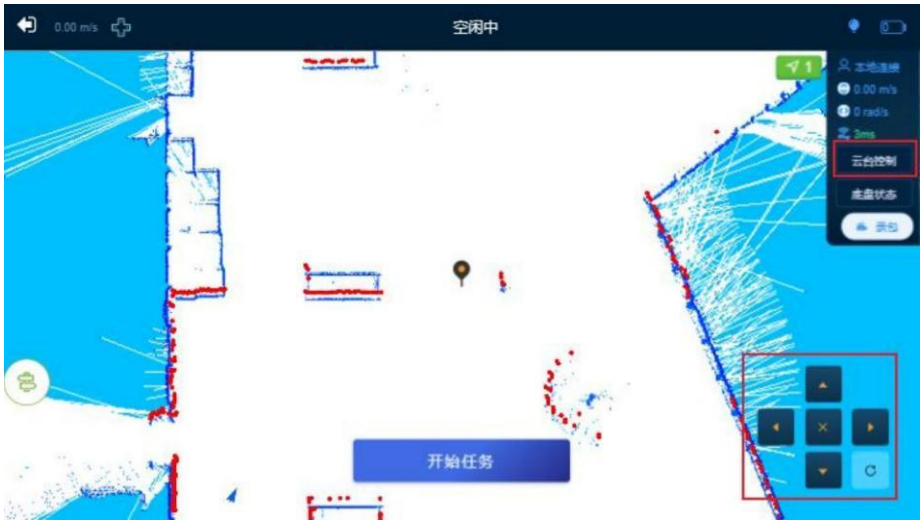

Fig 5-7

5.6 External Device Control

Horn, light, police light control:

In the APP home page (as in Figure 5-1), click the “ ‘ icon button to enter into the [Function Setting] interface, click the ’ ” icon in the upper right corner to expand the hidden setting area, and click the corresponding icon button to control the horn, light and police light switch. As Figure 5-2

5.7 Audio Collections

In the APP home page (as in Figure 5-1), click the “ ” icon button, select [Audio Favorites], you can view the list of uploaded audio, click [Play] to play the audio, you can play more than one audio at the same time, and click [Stop] in the upper right corner to stop all the playing sounds.

Upload Audio: Click [Upload] at the top right corner, select the audio in .wav format in the folder of your mobile device and upload it, the audio will be saved in the robot system after successful upload.

6 Operational processes

1. First, connect to the robot's WiFi via your tablet or cell phone.
2. Open the Robot Operation Platform APP, configure the robot name, IP, port, user name and password, and then click login. 3.

3. After logging in, enter the home page, click on the icon button to enter the [Select Map] function interface, click on the upper right corner of the [Build Map] to jump to the [Create Map] function interface.
4. Click [start building], click the interface above the " " icon, pop-up remote control interface, the remote control robot to control the movement of the laser swept to the environment will be displayed on the right side of the completion of the construction of the map, click [Save], pop-up box, for the map to name a name, click [OK], a map will be completed. The precautions for building a map have been described above.
5. After the completion of the construction of the map, the first to add a virtual wall to the map: home page, click back to the icon button , go to [select map] you can add a virtual wall.
6. Position initialization, to help the robot correctly positioned in the map, only after accurate positioning can be correctly navigated; Steps: Click the " " icon on the home page to enter the [Position Initialization] main interface for the robot's position initialization operation.
7. Navigation

- Execution of multiple cruise points (recommended)

1) Record Points: Hand-drawn Points or Record Points function to add desired cruise points;

2) Drawing Tracks: Outdoor environments or large venues require cruising tasks to be accomplished via tracks;

3) Create multi-point navigation tasks;

4) Implementation of implementation mandates

- Execution follows the path

1)Recorded paths: done by recording paths or hand-drawn paths;

2)Creating follow-the-path tasks

3)Implementation of implementation mandates

7 Common Failure Analysis

problematic phenomenon	Possible causes	countermeasures
Builds with ghosting or builds that don't match reality	Laser, incorrect controller placement angle	Laser placed horizontally and angled to face forward Controller horizontal and angled to face forward
	No loopbacks in the build path	Rebuild the map in strict accordance with the rules for building maps
Firmware upgrade failure	Internet instability	Re-upgrade after reboot
	Shutting down the software or machine during an upgrade	
Suddenly stops and does not run while traveling	Insufficient power supply to the robot	Please recharge in time
	The motor is offline.	Recheck the motor wiring and motor status, make sure it is normal and restart it
	Laser dropout	Check and securely install the laser wiring to ensure proper functioning and reboot.
Stuck backward while traveling	Obstruction in front of the laser Check the ultrasound for obstructions and remove them	Check ultrasound for obstructions and remove
	ultrasound interference	Check the working status of the ultrasonic waves and adjust the ultrasonic wave measuring distance parameter appropriately, recommended 20-40cm; turn on and off the appropriate number of ultrasonic waves; the distance between two ultrasonic waves should not be too close.
	infrared interference	Check the infrared working status, detect whether the infrared device is placed horizontally and whether it detects the floor
Sudden turnaround of the barricade while traveling to the barricade	Current road path congestion	See if the current travel path is blocked or has less room to maneuver.
The machine oscillates from side to side during driving.	Laser placement is shifted	Reinstallation of correct laser position and orientation
	The new environment is highly variable and deviates significantly from the maps being implemented	New Map

	Creating map ghosting	Rebuild the map in strict accordance with the rules for building maps